

Hochschild Homology:

classical, topological, & motivic

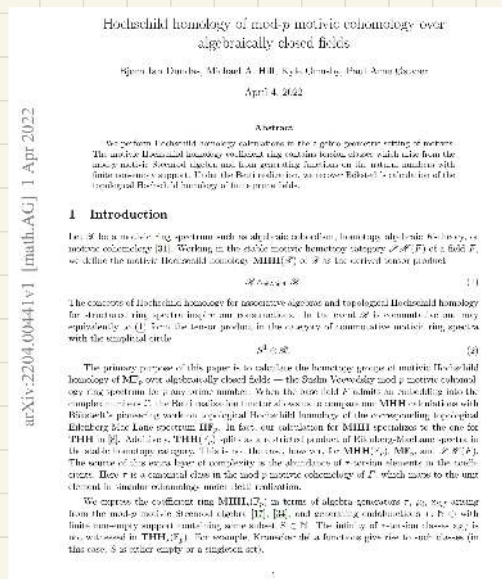


Homotopy
Arithmetic → Geometry
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30.VI.22

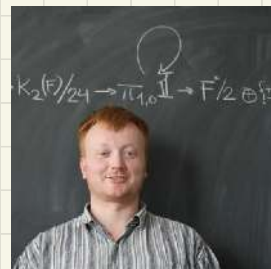


Outline

- ① ~~Hochschild homology~~
- ② Topological & motivic variants
- ③ Computations over $\text{Spec } \mathbb{C}$
- ④ Two truths and a lie?



All work joint with Bjørn Dundas, Mike Hill, & Paul Arne Østvær



Topological Hochschild Homology

- For a ring spectrum R , $\mathrm{THH}(R) := R \wedge_{R \wedge R^{\mathrm{op}}} R$.
- If A is a classical commutative ring, $\mathrm{THH}(A) := \mathrm{THH}(HA)$
Eilenberg-MacLane spectrum
- When R is E_∞ , we have

$$\mathrm{THH}(R) = S' \otimes R$$

spectra are tensored over simplicial sets

arising from

$$S' = \operatorname{colim} \left(\begin{array}{c} S^0 \\ \downarrow \\ I \end{array} \longrightarrow * \right).$$

Computations

Since $THH(A) = HA \wedge_{HA \wedge HA} HA$ and $\pi_* HA = A$ (in deg 0),

we have a Tor-spectral sequence

$$E_{h,t}^2 = \text{Tor}_{h,t}^{\pi_* HA \wedge HA} (A, A) \Rightarrow THH_{h+t}(A)$$

$$\text{and } d^r : E_{h,t}^r \rightarrow E_{h-r, t+r-1}^r.$$

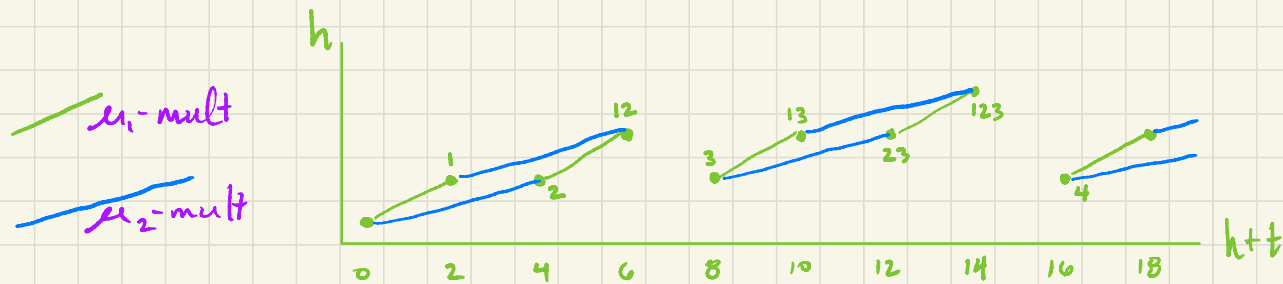
For $A = \mathbb{F}_2$, $\pi_* H\mathbb{F}_2 \wedge H\mathbb{F}_2 = \mathcal{A}_* = \mathbb{F}_2[\xi_1, \xi_2, \xi_3, \dots]$, $|\xi_i| = 2^i - 1$
is the mod-2 dual Steenrod algebra.

Thus $E^2 \cong \bigwedge_{\mathbb{F}_2} (\mu_1, \mu_2, \mu_3, \dots)$ with $|\mu_i| = (1, 2^i - 1)$.

Pictorially ...

Computations (ct'd)

$$E^2 \cong \bigwedge_{\mathbb{F}_2} (\mu_1, \mu_2, \mu_3, \dots) \quad \text{with } |\mu_i| = (1, 2^i - 1)$$



Note ① One class in each even degree.

② Differentials go 1 left & down, so $E^2 = E^\infty$.

Fact By a power operations argument, $\mu_i^2 = \mu_{i+1} \in THH_*(\mathbb{F}_2)$.

Thm (Bökstedt) $THH_*(\mathbb{F}_p) = \mathbb{F}_p[\mu]$, $|\mu| = 2$ for all primes p .

<new>

Motivic Hochschild Homology

Now move from spectra to motivic spectra and replace HA with MA , the motivic Eilenberg-Mac Lane spectrum.

Morel-Voevodsky:

$$X \in \mathcal{A}' \rightarrow Y$$

$[Sm/k^{\text{op}}, [\Delta^{\text{op}}, \text{Set}]] + \text{Nisnevich \& } \mathcal{A}'\text{-localize} = \text{motivic spaces } Spc_k$

$$\left(\begin{array}{ccc} & \nearrow \gamma & \\ Sm/k & & [\Delta^{\text{op}}, \text{Set}] \\ & \nwarrow \text{const} & \end{array} \right)$$

invert $\mathbb{P}' \wedge -$

motivic spectra Sp_k

Motivic Hochschild Homology (ct'd)

$$\begin{array}{ccc}
 A' \setminus 0 & \longrightarrow & A' \simeq * \\
 \downarrow \quad \uparrow & & \downarrow \\
 * \simeq A' & \longrightarrow & P'
 \end{array}
 \quad \Rightarrow \quad P' \simeq S' \wedge (A' \setminus 0) .$$

simplicial circle
geometric circle

Upshot Bigraded spheres $S^{m,n} := (S')^{m-n} \wedge (A' \setminus 0)^{\wedge n}$

total degree
weight

Thus we have bigraded homotopy groups

$$\pi_{m,n} X := [S^{m,n}, X]$$

for $X \in \mathcal{S}_k$.

Note Need homotopy sheaves to detect weak equivs.

Motivic Hochschild Homology (ct'd)

Important (co)homology theories are representable in Sp_k :

- MA = motivic cohomology, with coefficients in A
- KGL = (homotopy) algebraic K-theory
- KQ = Hermitian K-theory
- MGL = algebraic cobordism

We define the motivic Hochschild homology of a commutative ring A to be the motivic spectrum

$$MHH(A) := MA \wedge_{MA \wedge MA} MA = S^1 \otimes MA$$

$$\dots \circ \left\{ (A^{\wedge 0}) \otimes MA \right.$$

with coefficients $MHH_{*,*}(A) = \bigoplus_{m,n \in \mathbb{Z}} \pi_{m,n} MHH(A)$.

Computations over $\mathbb{C}$

Fix k algebraically closed, $A = \mathbb{F}_p$, $p \neq \text{char}(k)$. $\begin{cases} \text{THH}_*(\mathbb{F}_p) \\ = \mathbb{F}_p[\mu] \end{cases}$

Theorem 1.1. Over an algebraically closed field of exponential characteristic $e(F) \neq p$, there is an algebra isomorphism

$$\mathbf{MHH}_*(\mathbb{F}_p) \cong \mathbb{F}_p[\tau, \mu_i, x_{S,f}]_{i \in \mathbb{N}, (S \subset \text{supp } f, f: \mathbb{N} \rightarrow \mathbb{N})} / \mathcal{I} \quad (3)$$

with the ideal of relations

$$\mathcal{I} = \left(\begin{array}{c} \mu_i^p - \tau^{p-1} \mu_{i+1}, \\ \tau^{p-1} x_{S,f}, \\ x_{S,f} \cdot x_{T,g} - \sum_{u \in \text{supp}(f+g) - S \cup T} \epsilon_u \cdot x_{S \cup T \cup \{u\}, f+g} \end{array} \right).$$

'finite support'

Here the support of f is a finite non-empty subset of the natural numbers and $S \subset \text{supp } f \subset \mathbb{N}$ does not contain the minimal element of $\text{supp } f$. The coefficient $\epsilon_u \in \mathbb{F}_p$ is given explicitly in Definition 2.12. The algebra generators have bidegrees given by $|\tau| = (0, -1)$, $|\mu_i| = (2p^i, p^i - 1)$, and

$$|x_{S,f}| = (|S| + 1)(-1, p - 1) + p \sum_{j \in \text{supp } f} f(j)(2p^j, p^j - 1).$$

Computations over $\mathbb{C}$ (ct'd)

Fix k algebraically closed, $A = \mathbb{F}_p$, $p \neq \text{char}(k)$.

Theorem 1.1. Over an algebraically closed field of exponential characteristic $e(F) \neq p$, there is an algebra isomorphism

$$\text{MHH}_*(\mathbb{F}_p) \cong \mathbb{F}_p[T, \mu_i, x_{S,f}]_{i \in \mathbb{N}, (S \subset \text{supp } f, f: \mathbb{N} \rightarrow \mathbb{Z})} \quad (3)$$

with the ideal of relations

$$T = \left(x_{S,f} \cdot x_{T,g} - \sum_{u \in \text{supp}(f+g) - S \cup T} \mu_u \cdot x_{S, T \cup \{u\}, f+g} \right),$$

Here the support of f is a finite non-empty subset of the natural numbers and $S \subset \text{supp } f \subset \mathbb{N}$ does not contain the minimal element of $\text{supp } f$. The coefficient $\mu_u \in \mathbb{F}_p$ is given explicitly in Definition 2.12. The algebra generators have bidegrees given by $|v| = (0, -1)$, $|\mu_i| = (2p^i, p^i - 1)$, and

$$|x_{S,f}| = (|S| + 1)(-1, p - 1) + p \sum_{j \in \text{supp } f} f(j)(2p^j, p^j - 1).$$

Definition 2.12. For functions $f, g: \mathbb{N} \rightarrow \mathbb{Z}$ with finite support and non-empty finite sets $S, T \subseteq \mathbb{N}$ define $K_{S,T,f,g} \in \mathbb{F}_p$ by

$$K_{S,T,f,g} = \left(\prod_{s \in S} \binom{fs - 1 + gs}{fs - 1} \right) \left(\prod_{t \in T} \binom{ft + gt - 1}{ft} \right) \left(\prod_{c \notin S \cup T} \binom{fc + gc}{fc} \right)$$

if $(S, f), (T, g) \in J$ and $S \cap T = \emptyset$, and set $K_{S,T,f,g} = 0$ otherwise. Moreover, we define

$$\epsilon_{u,S,T,f,g} = K_{S \cup \{u\}, T \cup \{t_{f+g}\}, f+g} + K_{S \cup \{t_{f+g}\}, T \cup \{u\}, f+g}.$$

Yikes! Goals ① Shape of the computation when $p=2$.
② Consequences in motivic homotopy.

Computation Strategy

Input $\pi_{**} \mathbf{MF}_2 = \mathbb{F}_2[\tau], |\tau| = (0, -1)$

$$\Lambda_{**} = \pi_{**} \mathbf{MF}_2 \wedge \mathbf{MF}_2 \cong \mathbb{F}_2[\tau, \xi_1, \xi_2, \dots, \tau_0, \tau_1, \dots] / (\tau_i^2 - \tau \xi_{i+1} \mid i \geq 0)$$

$$|\xi_i| = (2^{i+1} - 2, 2^i - 1), \quad |\tau_i| = (2^{i+1} - 1, 2^i - 1)$$

Step 1 Calculate étale motivic Hochschild homology

$$\mathrm{MHH}_{**}(\mathbb{F}_2)[\tau^{-1}] \cong \mathbb{F}_2[\tau^{\pm 1}, \mu_0, \mu_1, \dots] / (\mu_i^2 - \tau \mu_{i+1} \mid i \geq 0)$$

$$\cong \mathrm{THH}_*(\mathbb{F}_2)[\tau^{\pm 1}].$$

Step 2 Calculate mod- τ MHH

$$\mathrm{MHH}_*(\mathbb{F}_2) / \tau \cong \bigwedge_{\substack{\mathbb{F}_2 \\ \text{divided powers algebra}}}(\bar{\mu}_0, \bar{\mu}_1, \dots) \otimes \bigwedge_{\mathbb{F}_2}(\bar{\lambda}_1, \bar{\lambda}_2, \dots).$$

Computation Strategy

Step 3 τ -torsion in $MHH_{**}(\mathbb{F}_2)$ injects into $MHH_{**}(\mathbb{F}_2)/\tau$ with image that of the τ -Bockstein.

Step 4 Give a presentation of τ -torsion in $MHH_{**}(\mathbb{F}_2)$ in terms of generators $x_{S,f}$ where $f: \mathbb{N} \rightarrow \mathbb{N}$ has finite support and $S \subseteq \text{supp } f$.

Step 5 Combine étale and τ -torsion computations via a pullback square.

Step 1 Étale MHH

$$\begin{aligned} \pi_{**} MF_2 \wedge MF_2[\tau^{\pm 1}] &\cong \mathbb{F}_2[\tau^{\pm 1}, \xi_1, \xi_2, \dots, \tau_0, \tau_1, \dots] / (\tau_i^2 - \tau_{i+1} \xi_i \mid i \geq 0) \\ &\cong \mathbb{F}_2[\tau^{\pm 1}, \tau_0, \tau_1, \dots], \quad |\tau_i| = (2^{i+1}-1, 2^i-1) \end{aligned}$$

so the Tor-spectral sequence takes the form

$$\begin{aligned} E^2 &= \text{Tor}_{*,*,*}^{\mathbb{F}_2[\tau^{\pm 1}, \tau_0, \tau_1, \dots]}(\mathbb{F}_2[\tau^{\pm 1}], \mathbb{F}_2[\tau^{\pm 1}]) \\ &\cong \bigwedge_{\mathbb{F}_2}(\mu_0, \mu_1, \dots)[\tau^{\pm 1}] \Rightarrow MHH_{**}(\mathbb{F}_2)[\tau^{\pm 1}] \end{aligned}$$

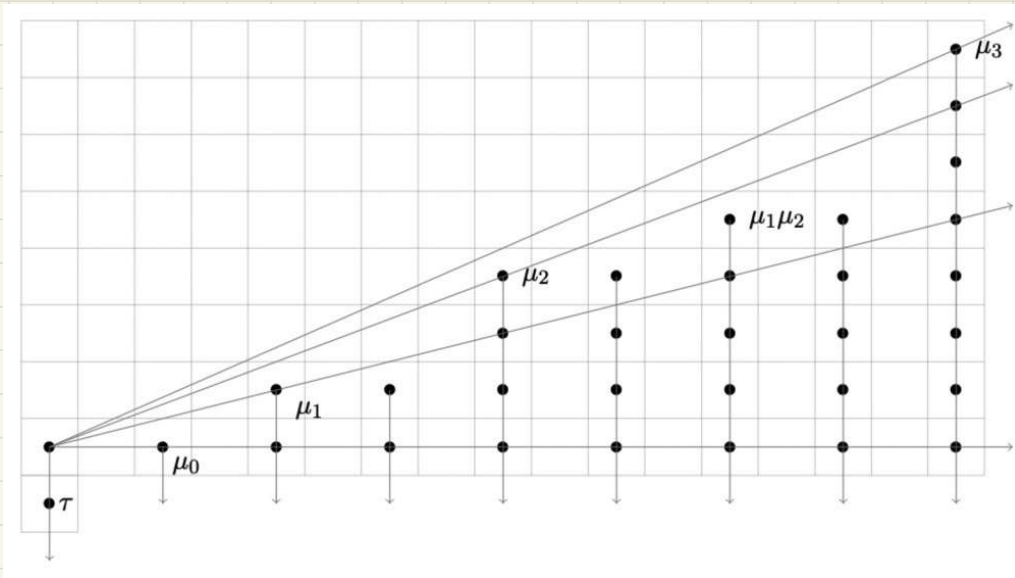
with $|\mu_i| = (1, 2^{i+1}-1, 2^i-1)$.

- Degree considerations $\Rightarrow E^2 = E^\infty$.
- Power operations $\Rightarrow \mu_i^2 = \tau \mu_{i+1}$.

Step 1 Étale MHH (ct'd)

$$\begin{aligned} \underline{U}_{\text{psht}} \text{ MHH}_{**}(\mathbb{F}_2)[\tau^{-1}] &\cong \mathbb{F}_2[\tau^{\pm 1}, \mu_0, \mu_1, \dots] / (\mu_i^2 - \tau \mu_{i+1} \mid i \geq 0) \\ &\cong \mathbb{F}_2[\tau^{\pm 1}, \mu_0] \end{aligned}$$

$$\begin{aligned} \text{im}(\text{MHH}_{**}(\mathbb{F}_2)) \\ \rightarrow \text{MHH}_{**}(\mathbb{F}_2)[\tau^{-1}] \end{aligned}$$



Step 2 Mod- τ MHH

$A_{**}/\tau \cong \mathbb{F}_2[\bar{\xi}_1, \bar{\xi}_2, \dots] \otimes \Lambda_{\mathbb{F}_2}(\tau_0, \tau_1, \dots)$ so the Tor-spectral sequence takes the form

$$\begin{aligned} E^2 &= \text{Tor}_{*,*,*}^{A_{**}/\tau}(\mathbb{F}_2, \mathbb{F}_2) \\ &\cong \Lambda_{\mathbb{F}_2}(\bar{\lambda}_1, \bar{\lambda}_2, \dots) \otimes \Gamma_{\mathbb{F}_2}(\bar{\mu}_0, \bar{\mu}_1, \dots) \Rightarrow \text{MHH}_{**}(\mathbb{F}_2)/\tau. \end{aligned}$$

divided powers algebra

Advanced degree yoga $\Rightarrow E^2 = E^\infty$ with no hidden ext's.

Upshot

$$\text{MHH}_{**}(\mathbb{F}_2)/\tau \cong \Lambda_{\mathbb{F}_2}(\bar{\lambda}_1, \bar{\lambda}_2, \dots) \otimes \Gamma_{\mathbb{F}_2}(\bar{\mu}_0, \bar{\mu}_1, \dots).$$

(See arXiv: 2204.0041 for Steps 3-5.)

④ Consequences

Since $THH(\mathbb{F}_p)$ is an $H\mathbb{F}_p$ -module, we get the splitting

$$THH(\mathbb{F}_p) \simeq \bigvee_{i \geq 0} \Sigma^{2i} H\mathbb{F}_p.$$

The τ -torsion in $MHH_{**}(\mathbb{F}_p)$ implies that this fails wildly in $Sp_{\mathbb{C}}$:

$MHH(\mathbb{F}_p)$ is not a free $M\mathbb{F}_p$ -module.

In Sp , the following are true:

- ① $H\mathbb{F}_2$ is a Thom spectrum of an E_2 -map with target $\Omega^2 S^3$
- ② $THH(\text{Thom}_1) = \text{Thom}_2 = \text{Thom}_1 \wedge B(\text{base}_1)_+$
- ③ $B\Omega^2 S^3 \simeq V(\text{spheres})$ stably $\eta_{\text{top}} = 0$

Consequences (ct'd)

Potential motivic analogues:

Behrens-Wilson: true
 C_2 -equivariantly

- ① MF_2 is a Thom spectrum over $\Omega^{2,1}S^{3,1}$.
- ② $MTH(Thom_1) \simeq Thom_1 \wedge B(base_1)_+$.
- ③ $B\Omega^{2,1}S^{3,1}$ is a wedge of spheres stably,
i.e. $\Sigma^\infty \Omega^{1,1}\Sigma^{1,1}S^2$ satisfies "Gru-James splitting".

Thm (Dundas, Hill, O, Østvær) At most two of these are true!

Consequences (ct'd)

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Poll Rank from most to least likely FALSE at
app.sli.do, code #MotivicHochschild.

Thank You!



Questions welcome.

Bonus content!

Power operations and $\mu_i^p = \tau^{p-1} \mu_{i+1}$:

- Mimic classical Dyer-Lashof operations via E_{∞} -structure.
- Show $Q^s \sigma = \sigma Q^s$: $\pi_{*} M\mathbb{F}_p \wedge M\mathbb{F}_p \longrightarrow MHH_{*+1+2s(p-1), p*}(\mathbb{F}_p)$ along classical lines. (Here $\mu_i = \sigma \tau_i$.)
- Rigidity + Betti realization $/\mathbb{C}$ imply the result, with τ^{p-1} compensating for weights.

Definition of $\chi_{S,f}$ for $f: \mathbb{N} \rightarrow \mathbb{N}$, $S \subseteq \text{supp}(f)$, $|\text{supp}(f)| < \infty$:

- $\chi_{S,f} = \tau$ -Bockstein of $\chi_{S,f}$ where
- $\chi_{S,f} = \left(\prod_{m \in S} \bar{\lambda}_{m+1} \gamma_{pf(m)-p} \bar{\mu}_m \right) \left(\prod_{n \notin S} \gamma_{pf(n)} \bar{\mu}_n \right).$